

Geometry Test 3 — Rigid Motions + Area/Perimeter

Mini-Review (read first!)

Distance (2D): $d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Perimeter/Area:

Rectangle: $P = 2(\ell + w)$, $A = \ell w$

Triangle: $A = \frac{1}{2}(\text{base})(\text{height})$

Circle: $C = 2\pi r$, $A = \pi r^2$

Midpoint: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Rigid motions about the origin:

Reflection in x-axis: $(x, y) \mapsto (x, -y)$

Translation by (a, b) : $(x, y) \mapsto (x + a, y + b)$

Rotation 90° CCW: $(x, y) \mapsto (-y, x)$

Rotation 180° CCW: $(x, y) \mapsto (-x, -y)$

Rotation 270° CCW: $(x, y) \mapsto (y, -x)$

Rigid motions preserve distances, perimeters, and areas.

1. **(Radius from a point)** A circle has center $O(2, -3)$ and passes through $P(5, 1)$. Find the radius using distance, then the circle's circumference and area (leave π).

2. **(Rectangle from coordinates)** Rectangle $ABCD$ has $A(1, 2)$, $B(9, 2)$, $C(9, -4)$, $D(1, -4)$. Use the distance formula to get ℓ and w , then compute perimeter and area.

3. **(Right triangle check)** $\triangle PQR$ with $P(0, 0)$, $Q(6, 0)$, $R(6, 8)$. (a) Verify right angle using distances. (b) Find perimeter and area.

4. **(Not axis-aligned base)** $\triangle ABC$ with $A(-3, 0)$, $B(3, 0)$, $C(0, 5)$. Use distance to find

each side length and the perimeter. Compute area using a convenient base/height.

5. (**Diameter given**) Diameter endpoints of a circle are $E(-4, 2)$ and $F(6, 2)$. Find the center, radius, circumference, and area.

6. (**Square via diagonal**) Opposite vertices of a square are $S_1(-2, 3)$ and $S_2(6, 11)$. Use the diagonal length to find the side length, then perimeter and area.

7. (**Triangle after reflection**) Reflect $\triangle A(3, 1)$, $B(7, 4)$, $C(1, 5)$ across the x-axis. List A' , B' , C' . What happens to perimeter and area?

8. (**Translate then rotate**) Point $P(-2, 3)$ is translated by $(a, b) = (5, -4)$ and then rotated 90° CCW about the origin. Find the final image P'' .

9. (**Rotate then translate**) Now start again at $P(-2, 3)$, first rotate 90° CCW, then

translate by $(5, -4)$. Compare with Q8: are the results the same? Explain briefly.

10. **(Preimage under reflection)** $A'(4, -7)$ is the image of A under reflection across the x-axis. Find A .

11. **(Find the translation)** A translation sends $M(-3, 2)$ to $M'(5, -1)$. (a) Find (a, b) . (b) Verify it also sends $N(0, 4)$ to N' ; give N' .

12. **(Which rotation?)** Which one of $90^\circ, 180^\circ, 270^\circ$ CCW rotations about the origin sends $U(4, -1)$ to $U'(-1, -4)$? Justify using the formulas.

13. **(Area preserved?)** $\triangle T$ has vertices $(0, 2), (4, 5), (8, 2)$. Compute its area. Then rotate it 180° CCW; what is the new area? (No coordinates needed if you argue correctly.)

14. **(Perimeter via distance)** Points $R(2, 1), S(9, 4), T(5, -2)$ form a triangle. Find all

three side lengths with distance and the perimeter (simplify radicals when reasonable).

15. (**Rectangle from three points**) $A(0,0)$, $B(8,0)$, $C(8,6)$ are consecutive vertices of a rectangle. (a) Find D . (b) Compute area and perimeter. (c) What happens to these if the rectangle is reflected across the x-axis?

16. (**Circle through a point**) A circle centered at the origin passes through $Q(12,5)$. Find the radius and the circle's circumference/area.

17. (**Two-step motion on a triangle**) $\triangle ABC$ has $A(1,2)$, $B(5,2)$, $C(5,6)$. Apply: translation by $(-3,1)$ then 270° CCW rotation. Give A'' , B'' , C'' (you may compute point by point).

18. (**Recovering a preimage**) After a 180° CCW rotation, a point becomes $(7, -9)$. What

was the original point?

19. (**Height via distance to a line (easy case)**) For $\triangle XYZ$ with $X(-4, 0)$, $Y(4, 0)$, $Z(1, 7)$, use base $\overline{XY}$ and a vertical height to find the area. Then find the perimeter.

20. (**Square inscribed in a circle**) A square is inscribed in a circle of radius $r = 10$. Find the square's side length and area. If you translate the square by $(6, -8)$, do these change?
